

JAVA
ADVANCED
INTERVIEW Q&A
PART – 2



@java_in.depth

ADVANCED JAVA INTERVIEW CONCEPTS: PART 2

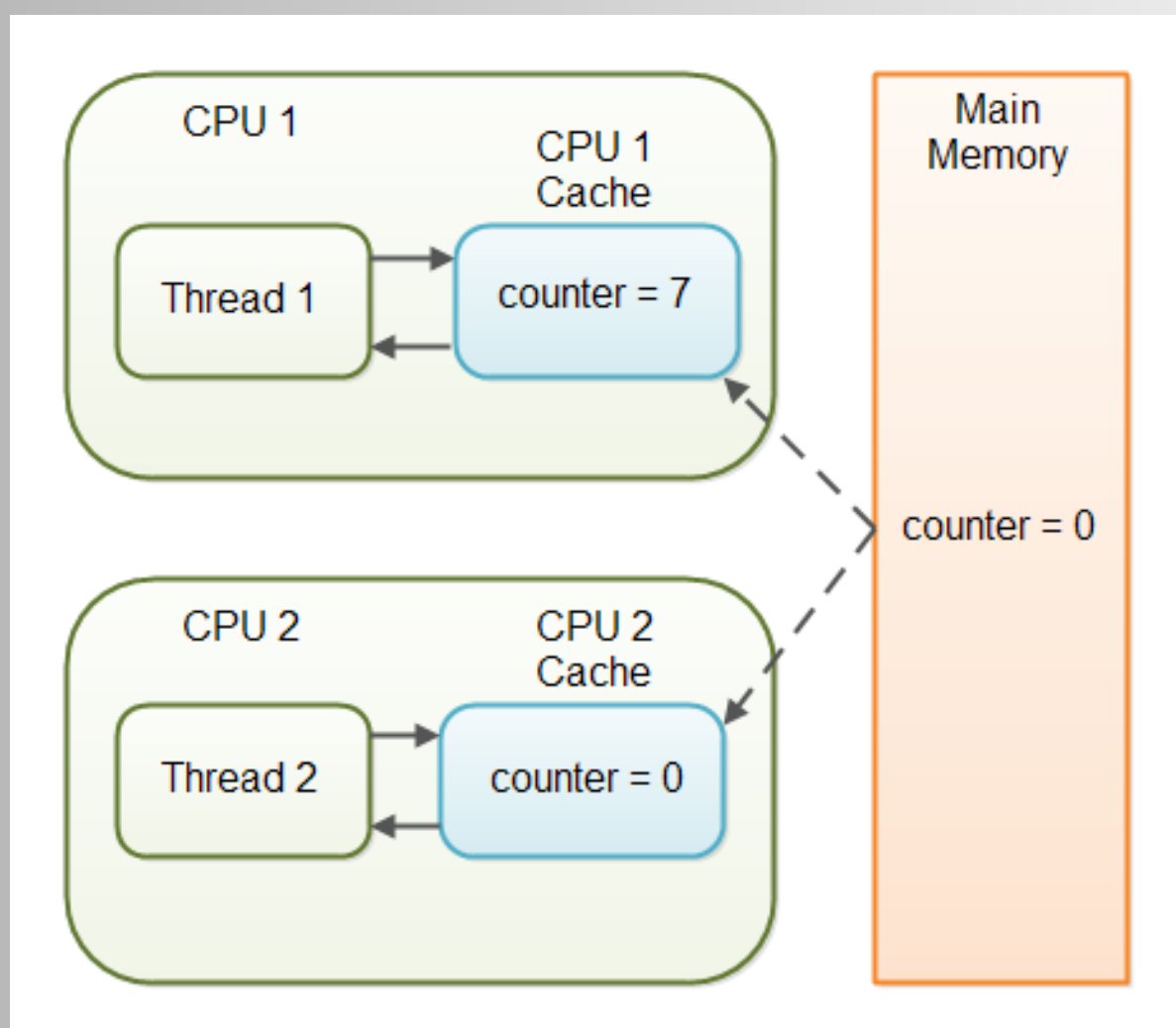
Concepts that interviewers ask at
mid/senior level

Save this post so you can revise these
concepts anytime.

1. WHAT IS VOLATILE KEYWORD?

volatile ensures that a variable's value is always read from main memory, not from a thread's local cache.

It guarantees visibility, but not full thread safety. It is useful when multiple threads read/write a variable but operations are simple and don't require locking.

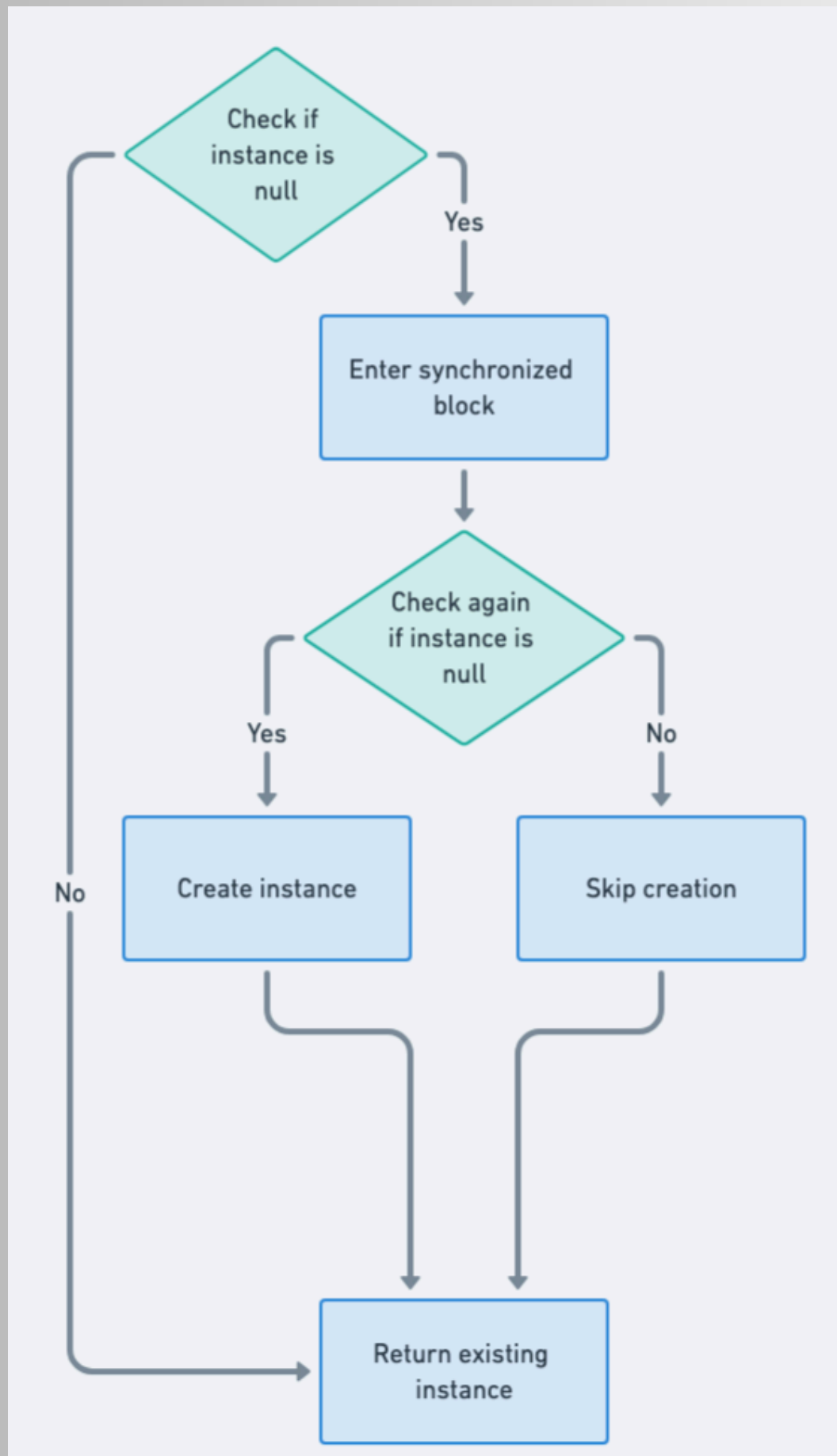


2. WHAT IS DOUBLE CHECKED LOCKING?

Double Checked Locking is used to create a thread-safe Singleton with better performance.

It avoids unnecessary synchronization by checking the instance twice. The variable must be volatile to prevent issues caused by instruction reordering.

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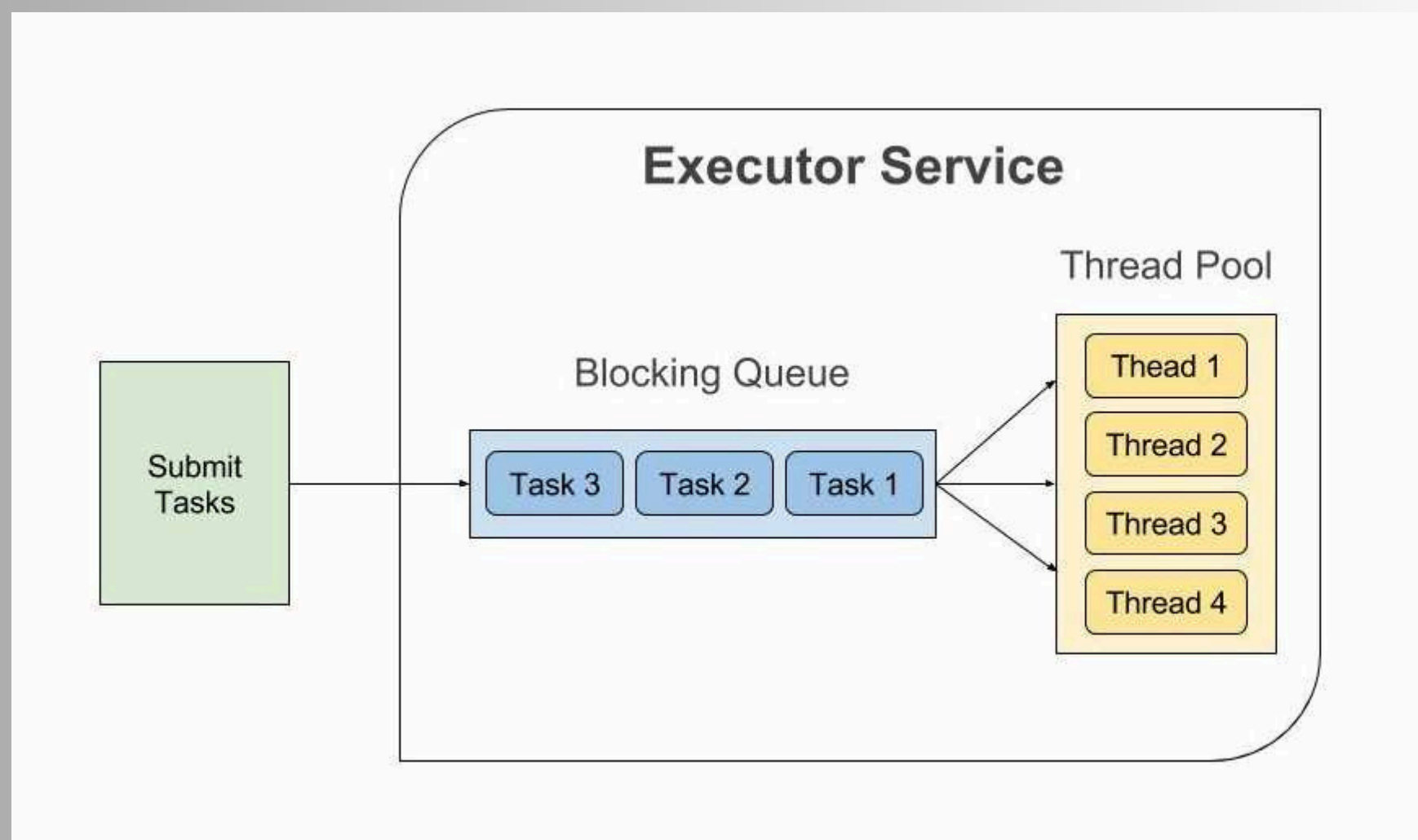


3. WHAT IS EXECUTORSERVICE?

ExecutorService manages a pool of threads instead of creating new threads manually.

It improves performance and resource management by reusing threads.

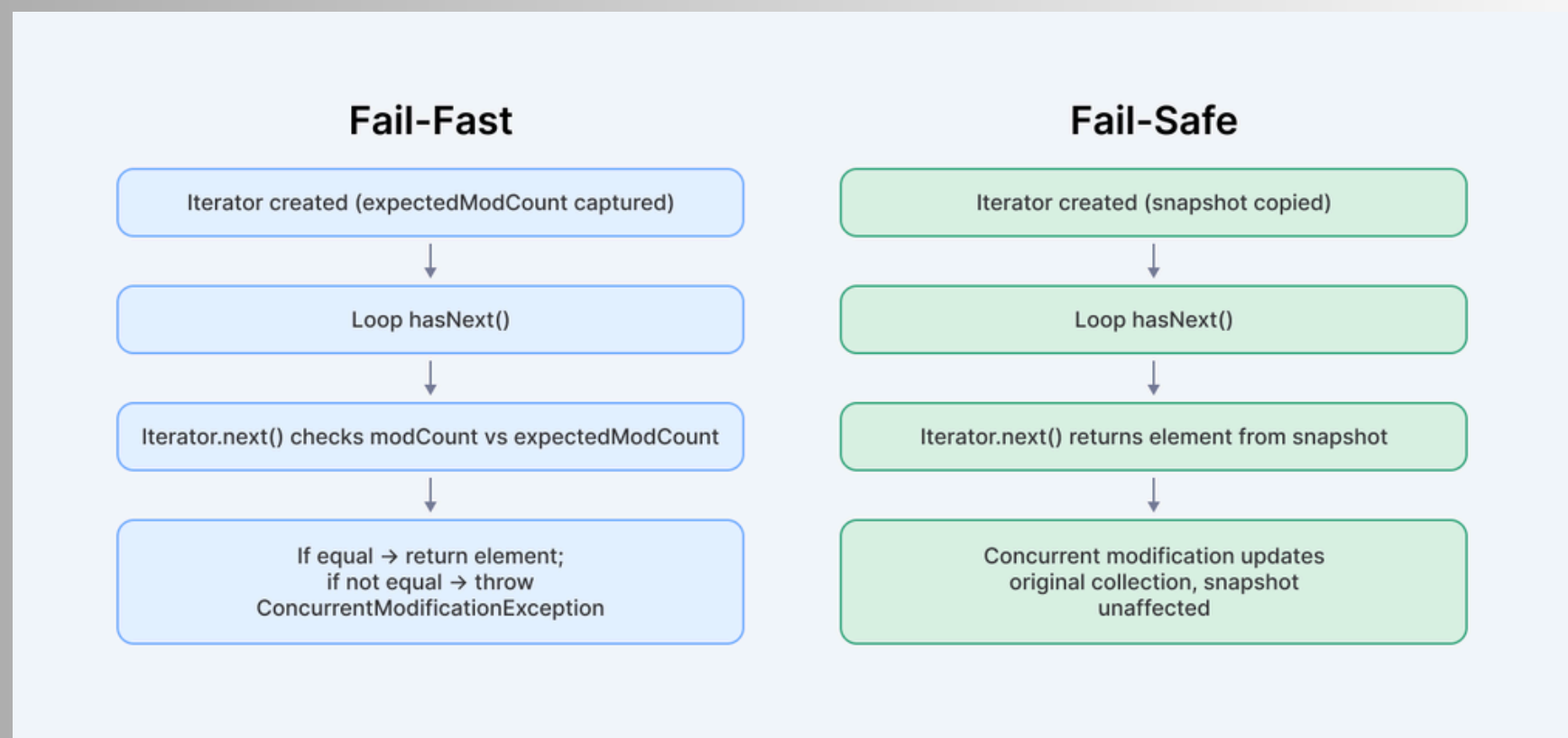
Common implementations include fixed thread pool and cached thread pool.



4. WHAT IS FAIL-FAST VS FAIL-SAFE ITERATOR?

Fail-fast iterators throw `ConcurrentModificationException` if the collection is modified during iteration.

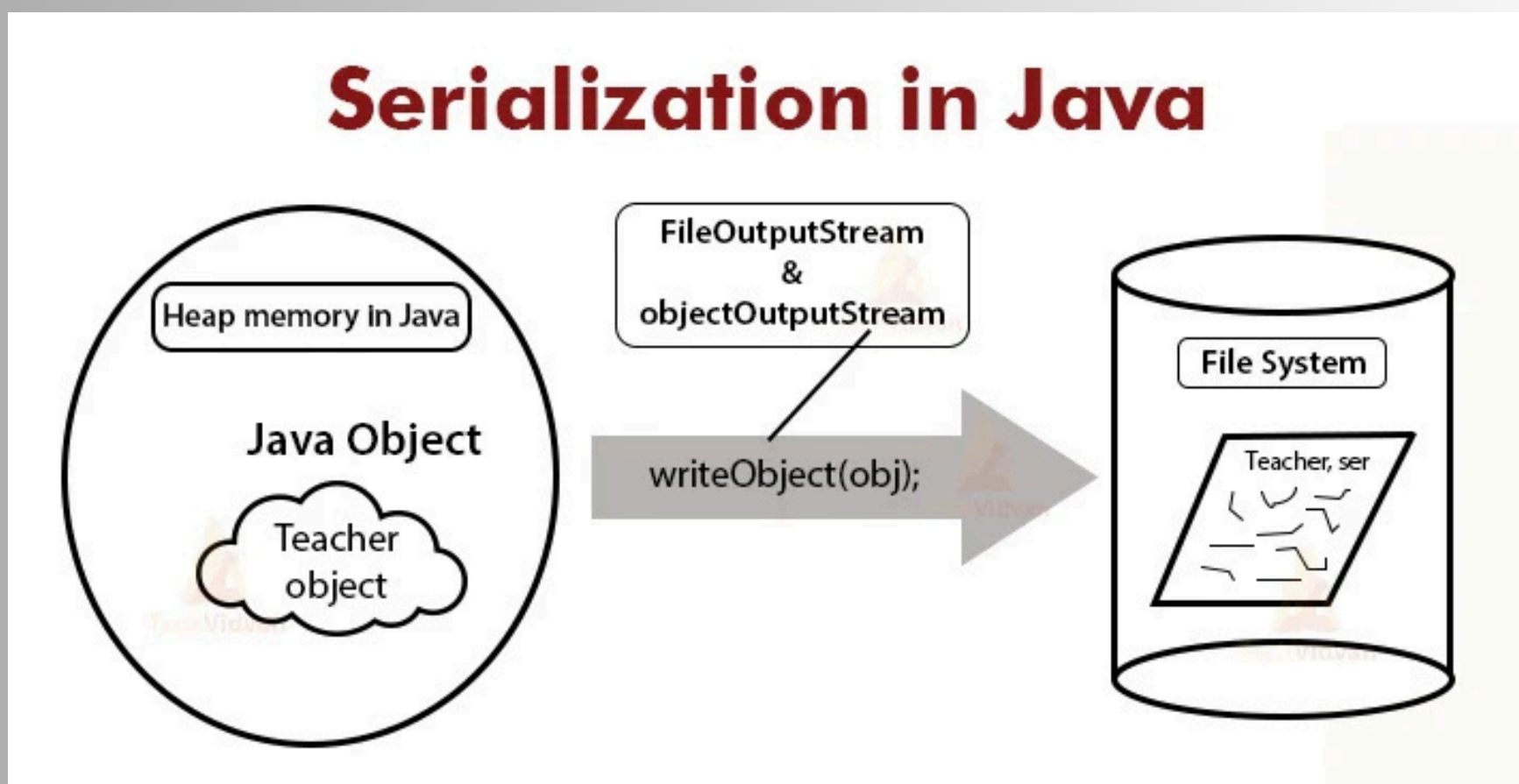
Fail-safe iterators work on a copy of the collection, so they don't throw exceptions but may not reflect real-time changes.



5. WHAT IS SERIALIZATION IN JAVA?

Serialization converts an object into a byte stream so it can be stored or transferred.

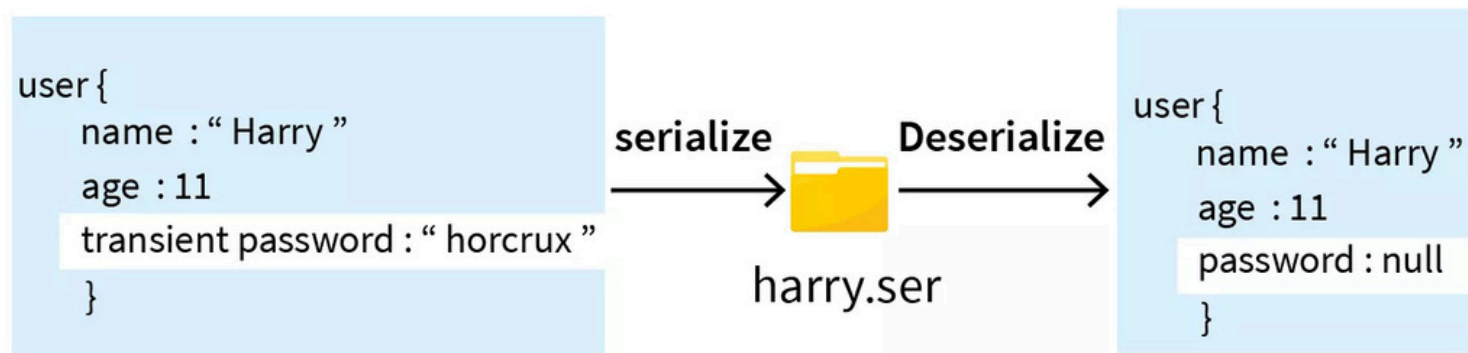
Deserialization reconstructs the object back. It is commonly used in caching, file storage, and network communication.



6. WHAT IS TRANSIENT KEYWORD?

transient is used to skip certain fields during serialization.

It is useful for sensitive data like passwords or fields that can be recalculated later.



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7. WHAT IS COMPARABLE VS COMPARATOR?

Comparable defines natural ordering inside the class using `compareTo()`.

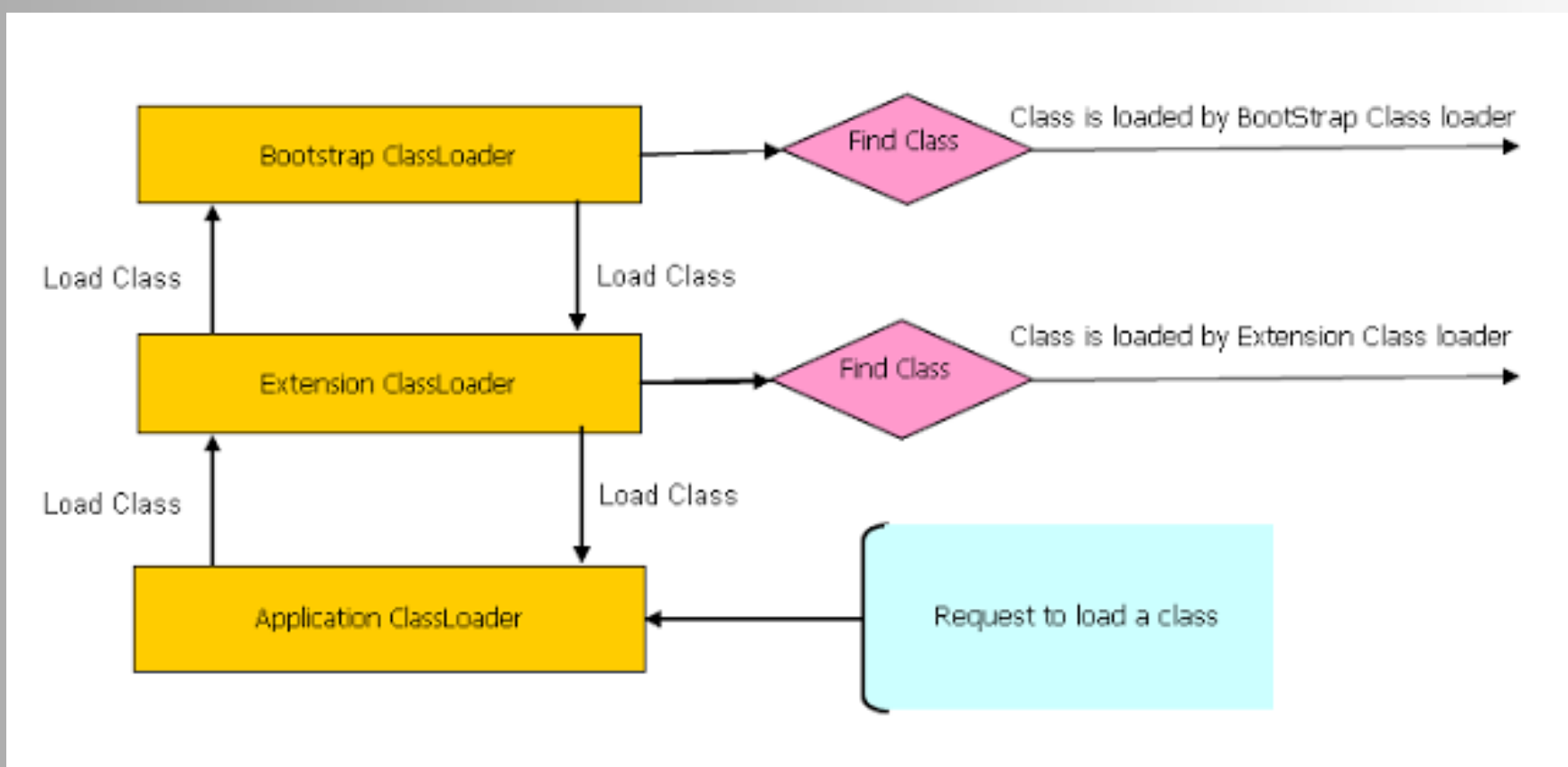
Comparator provides custom sorting logic outside the class using `compare()`.

Comparable in Java	Comparator in Java
Comparable interface is used to sort the objects with natural ordering.	Comparator in Java is used to sort attributes of different objects.
Comparable interface compares “this” reference with the object specified.	Comparator in Java compares two different class objects provided.
Comparable is present in <code>java.lang</code> package.	A Comparator is present in the <code>java.util</code> package.
Comparable affects the original class, i.e., the actual class is modified.	Comparator doesn't affect the original class
Comparable provides <code>compareTo()</code> method to sort elements.	Comparator provides <code>compare()</code> method, <code>equals()</code> method to sort elements.

8. WHAT IS CLASSLOADER?

ClassLoader loads Java classes into memory at runtime.

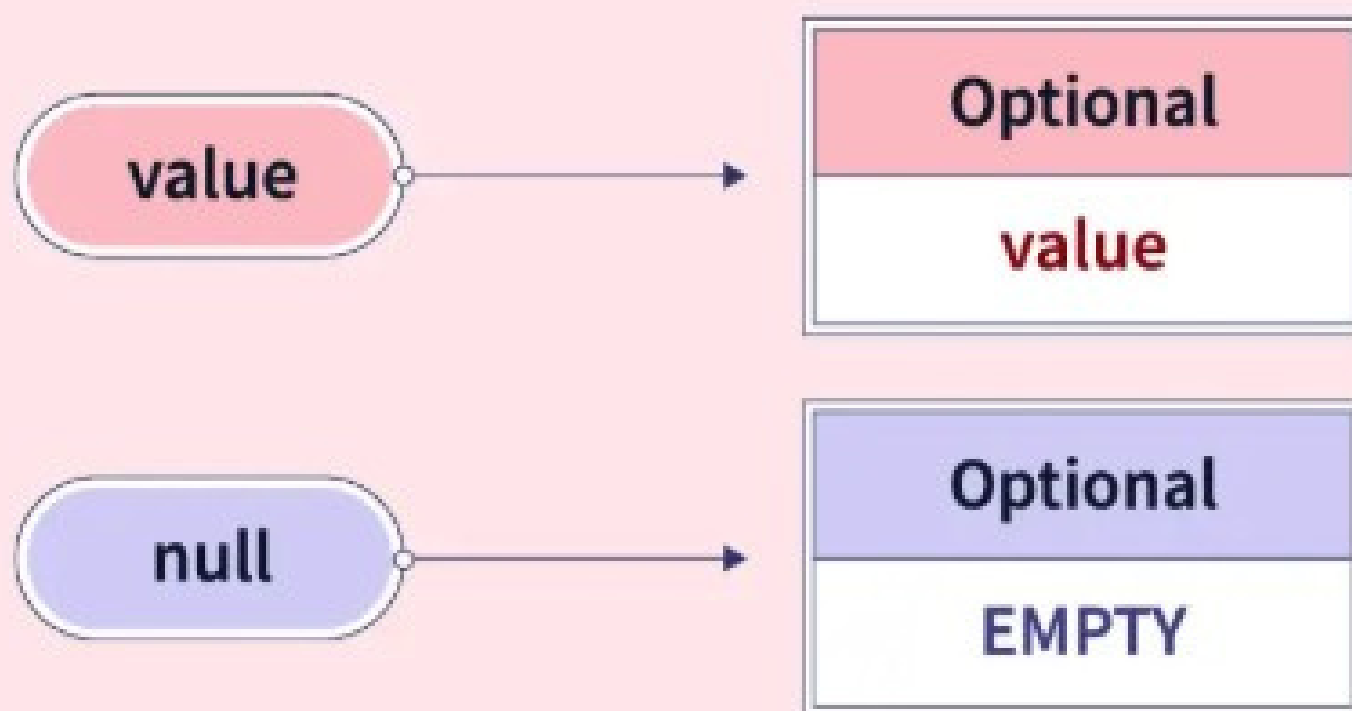
There are different types like Bootstrap, Extension, and Application ClassLoader. Understanding this helps in debugging and performance tuning.



9. WHAT IS OPTIONAL IN JAVA?

Optional is used to avoid `NullPointerException`.

It provides methods like `orElse()`, `ifPresent()`, and `map()` to handle missing values safely and cleanly.



WANT PART 3?

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